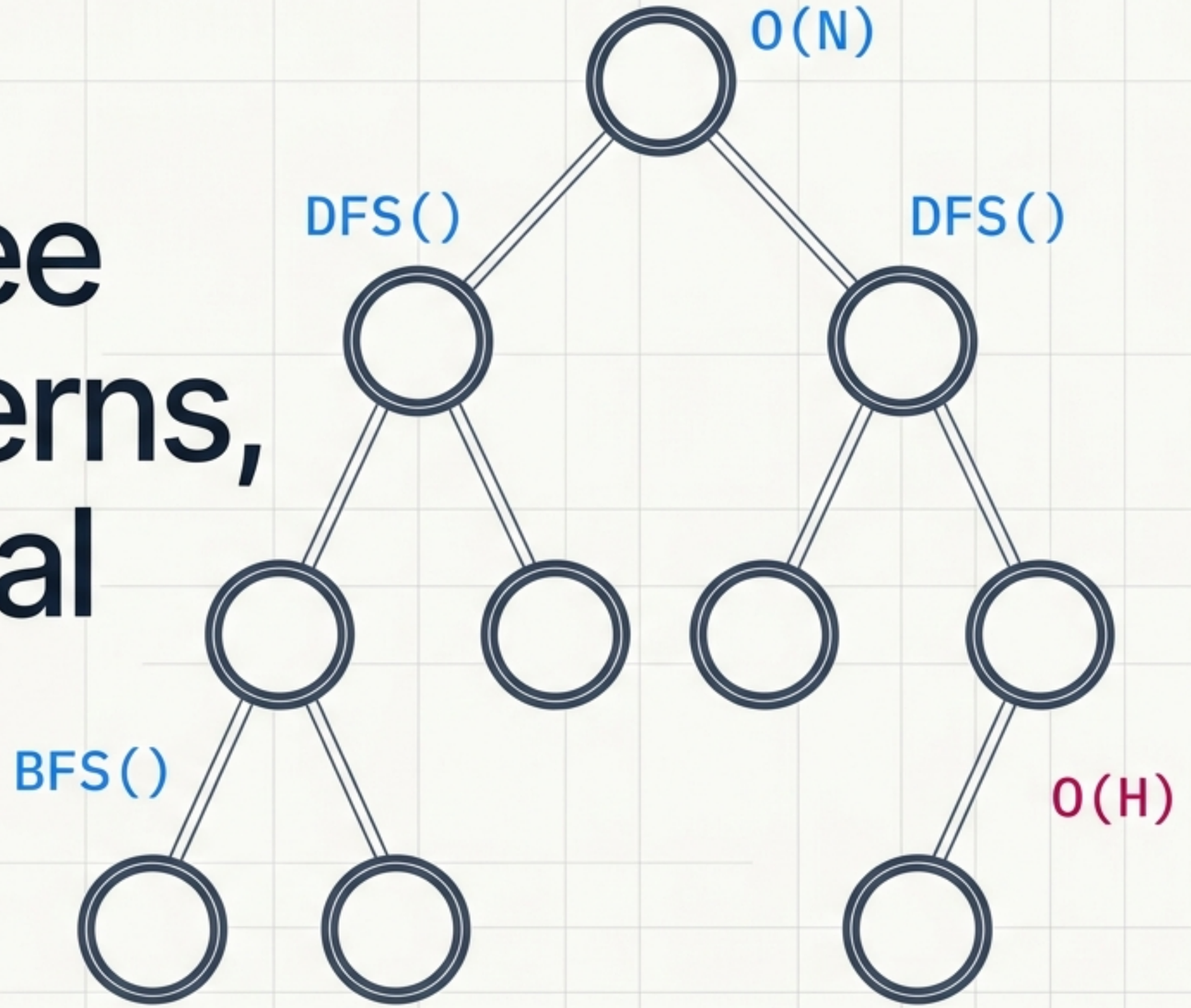
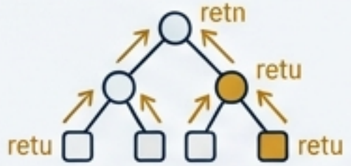
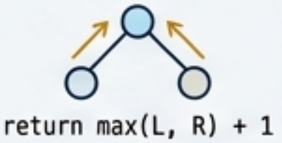
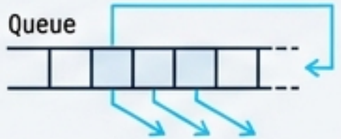
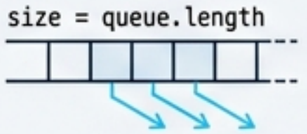
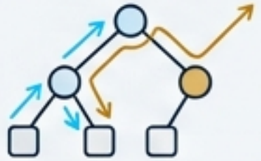
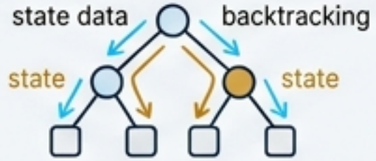
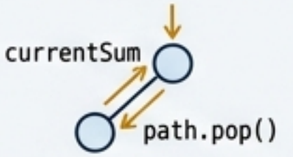
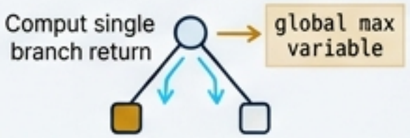
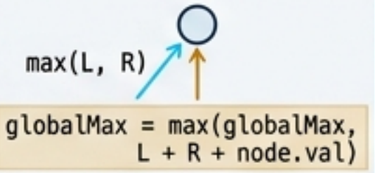
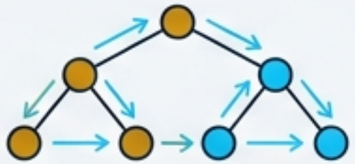
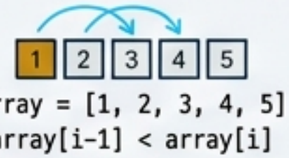
The FAANG Tree Playbook: Patterns, Visuals & Mental Models

High-yield blueprints for Google L4/L5 interview preparation.



Gemini Notebook

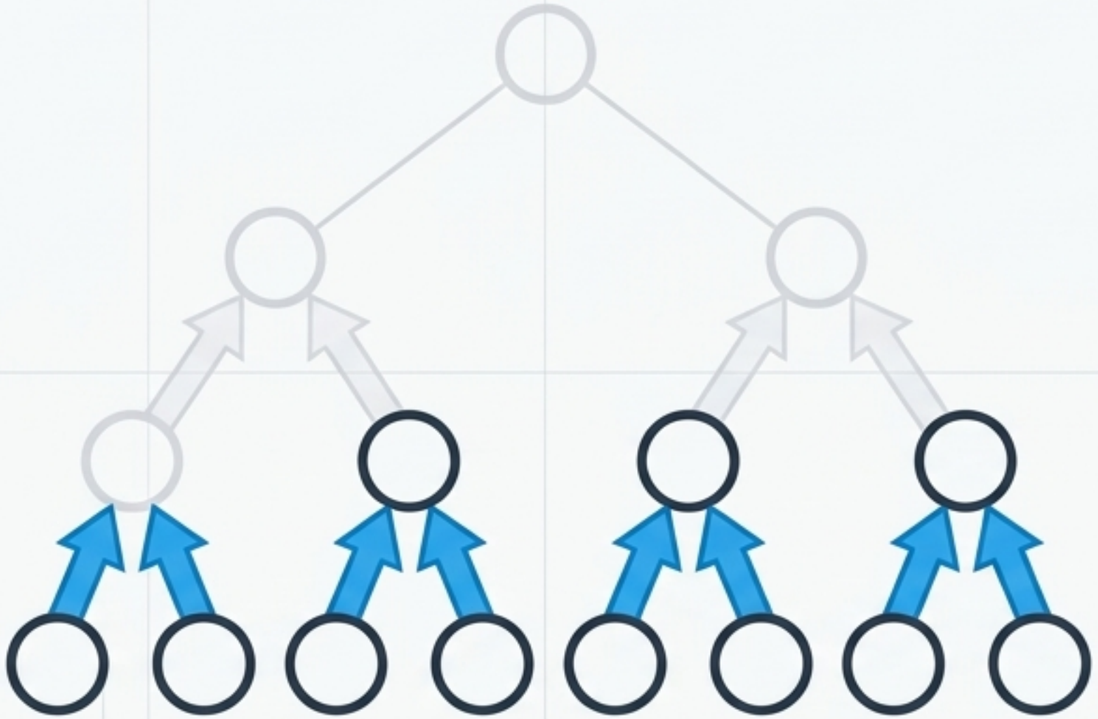
The Tree Pattern Diagnostic Matrix

If they ask for...	Think...	The Core Move
Depth, Height, Balance, Clone/Invert 	Bottom-Up DFS (Post-Order) 	Process children first; pass values up. 
Level Order, Views, Min Depth 	Queue BFS (Level-by-Level) 	Snapshot queue.length per level. 
Path Sums, Root-to-Leaf, Ancestors 	Top-Down DFS (Pre-Order) 	Pass state down; backtrack on return. 
Max Path Sum, Longest Path 	Global Max + Subtree Return 	Return single branch; update global with both. 
Sorted order, Validation, K-th Element 	BST In-Order Traversal 	Left -> Node -> Right = strictly increasing array. 

INVARIANT SPOTLIGHTING: Let the requested output shape dictate your traversal order.

Gemini Notebook

Bottom-Up DFS (The **Post-Order** Blueprint)



```
const left = dfs(node.left);    // 1. Process left
const right = dfs(node.right);  // 2. Process right

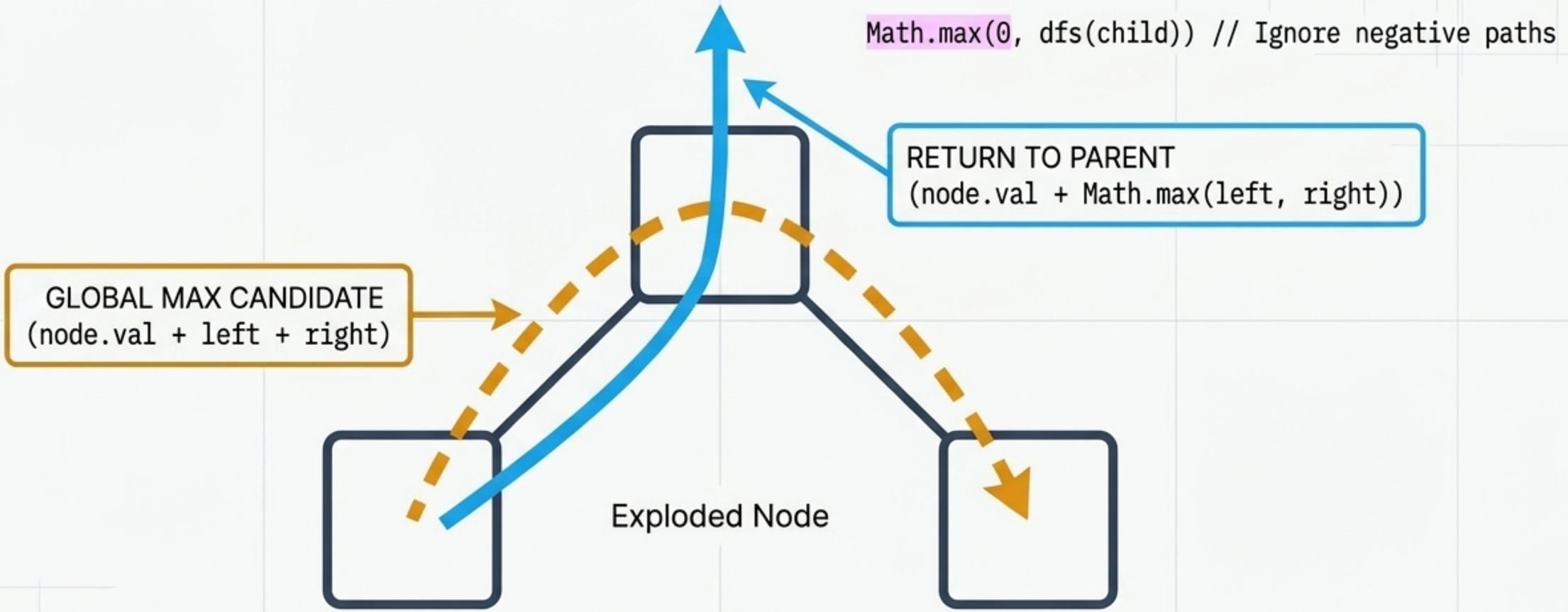
return 1 + Math.max(left, right); // 3. Aggregate
```

USE CASES: Height, Balance Check, Clone, Invert.

INVARIANT: A parent cannot compute its own value until its children return theirs. State bubbles up.

Gemini Notebook

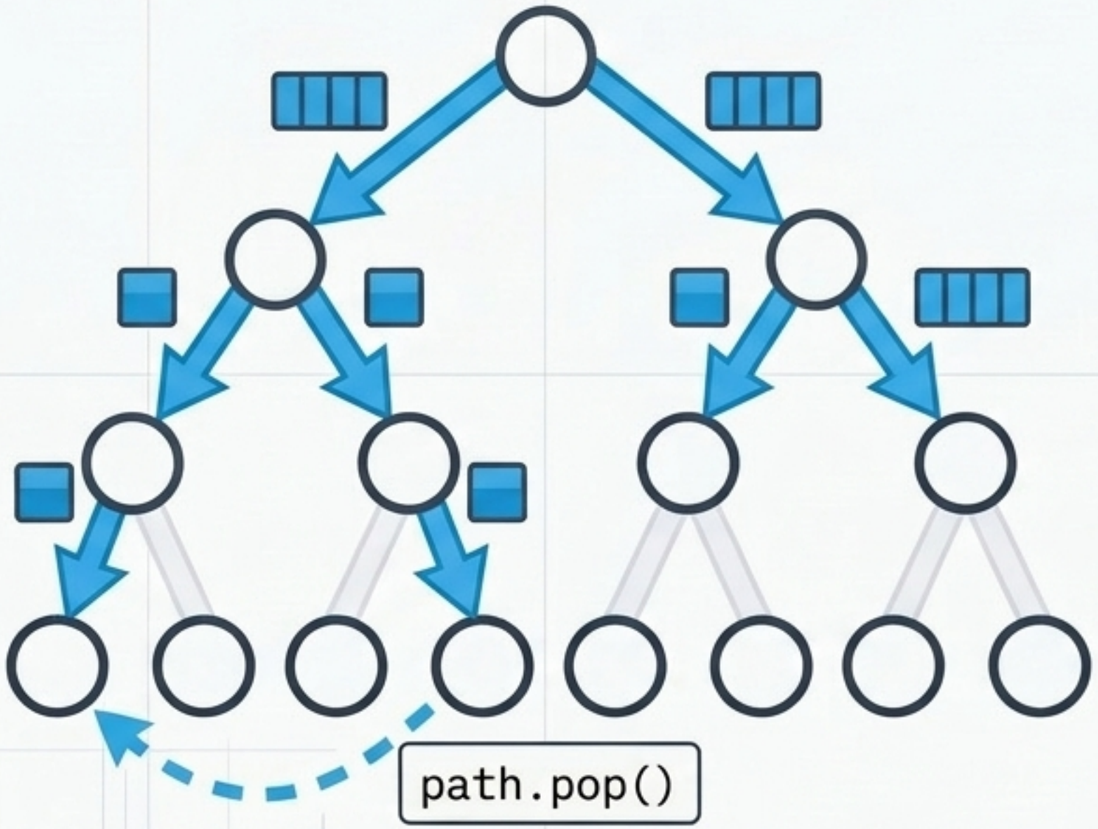
Spotlight: Binary Tree Maximum Path Sum (Hard)



INVARIANT: A path can curve across a node, but a parent node can only extend ONE upward branch.

Gemini Notebook

Top-Down DFS (The Pre-Order Blueprint)



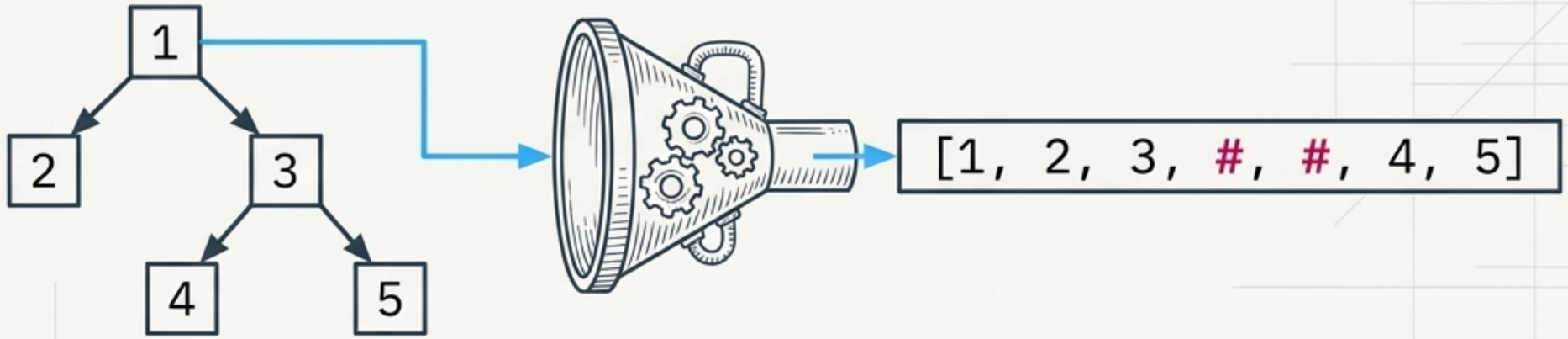

```
path.push(node.val);      // Choose
dfs(node.left, path);
dfs(node.right, path);
path.pop();               // Backtrack
```

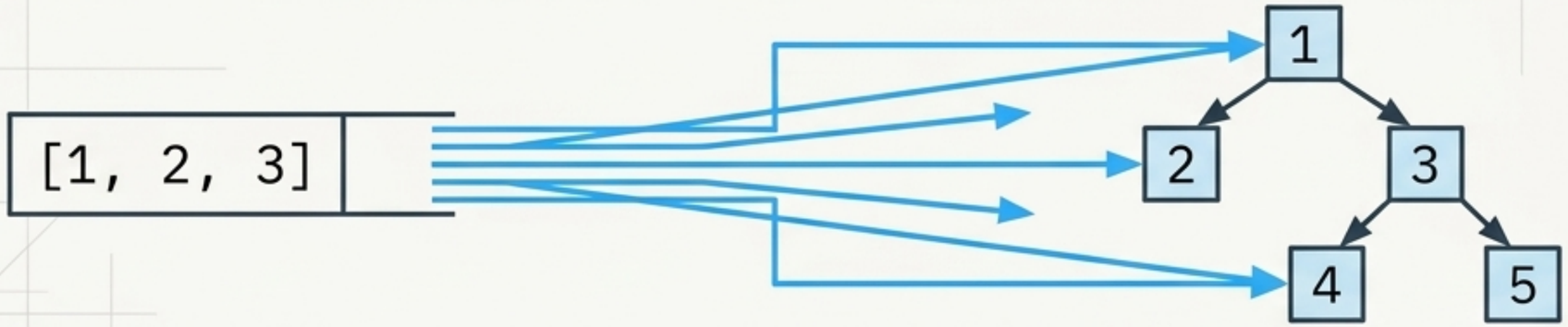
USE CASES: Path Sum I/II/III, Root-to-Leaf paths.

INVARIANT: State passed by **reference** must be **cleaned up** (popped) when retreating.

Gemini Notebook

Spotlight: Tree Serialization & Deserialization



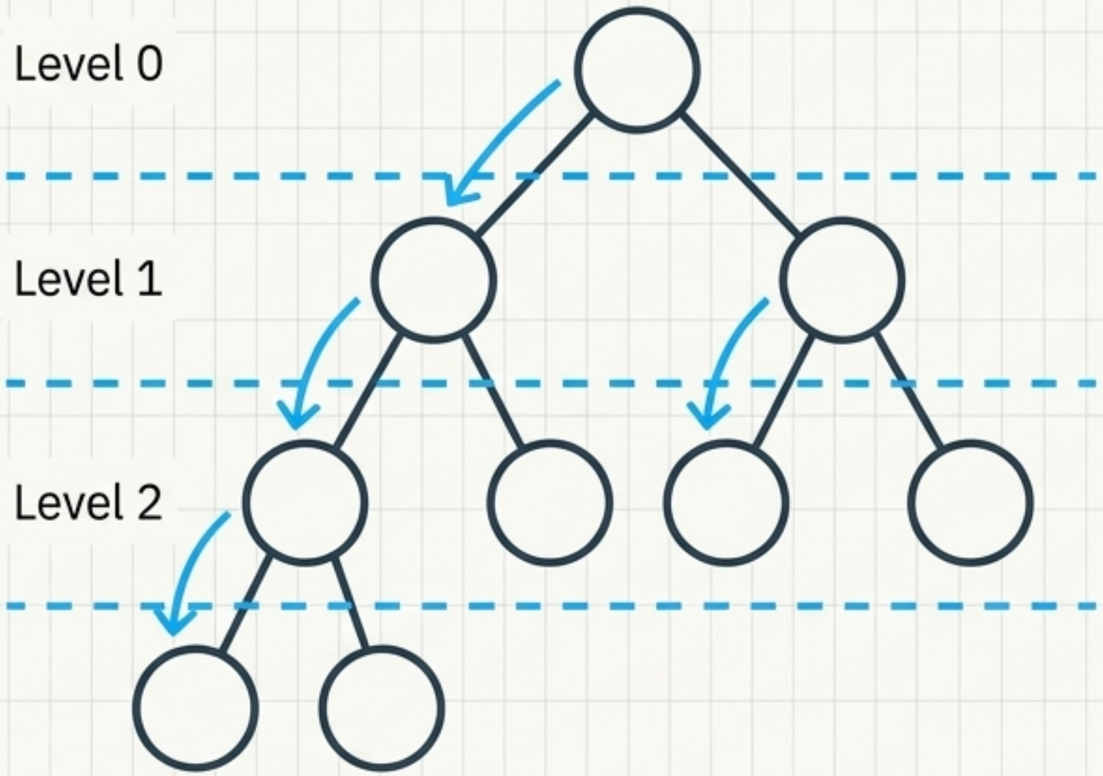


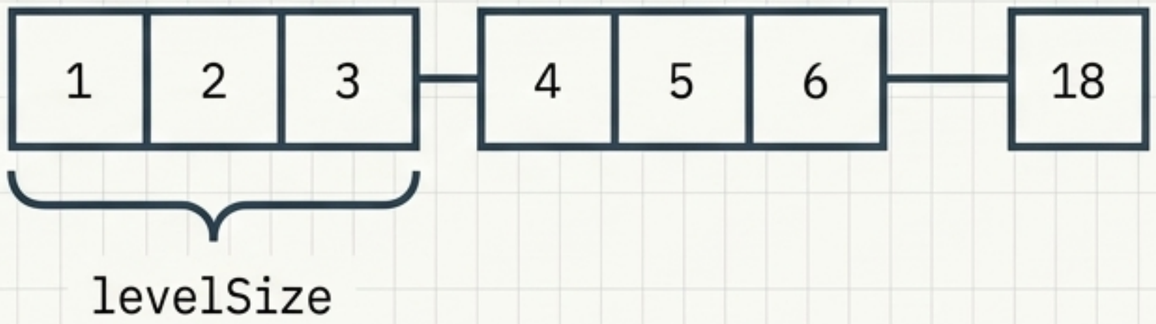
INVARIANT: Explicit null markers ('#') preserve structural boundaries. Without them, structural fidelity is lost during flattening.

- Works equally well with Queue (Level-Order) or Pre-Order DFS.

Gemini Notebook

Level-By-Level BFS (The Layered Blueprint)





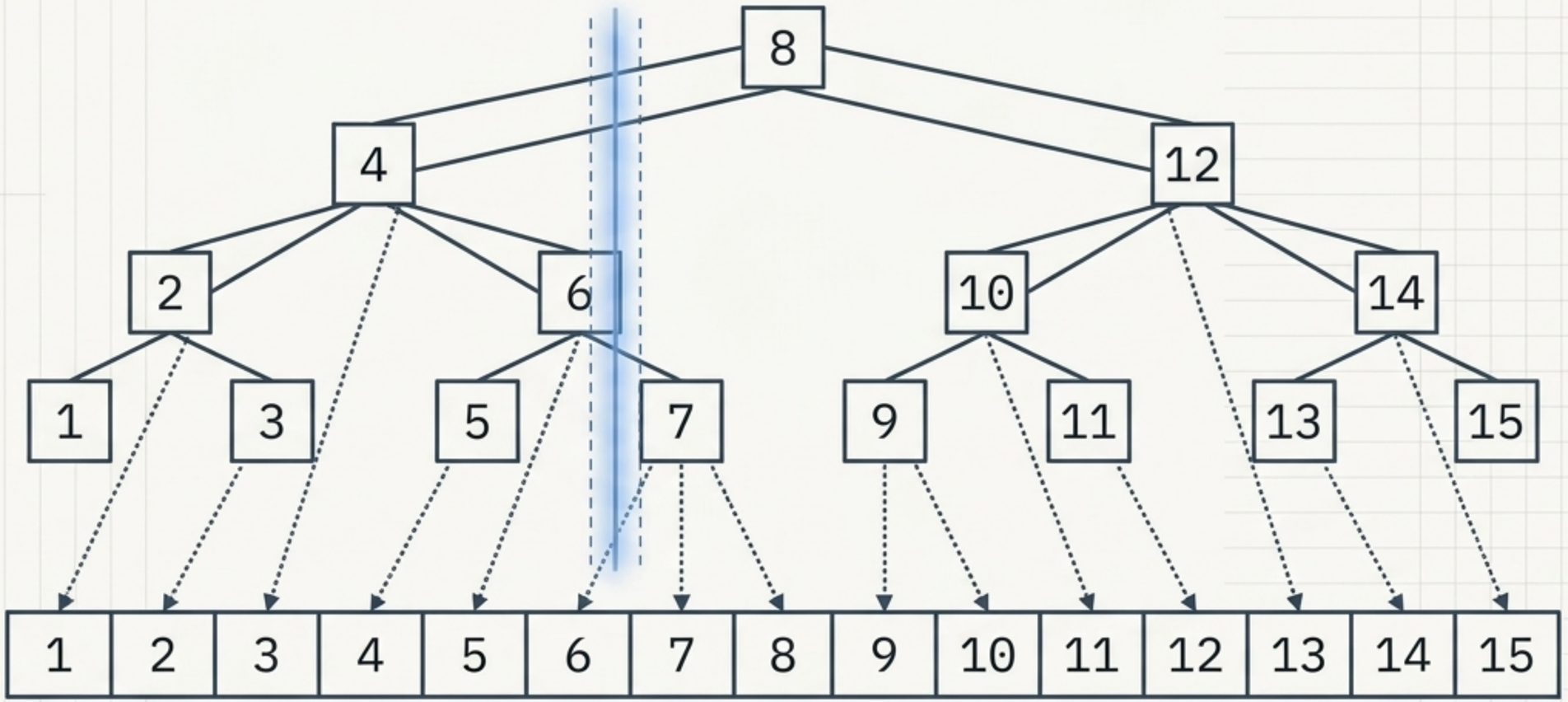

```
const levelSize = queue.length; // Snapshot
for (let i = 0; i < levelSize; i++) {
  const node = queue.shift();
  // push children
}
```

USE CASES: Right/Left side views, Minimum depth.

INVARIANT: Snapshot `queue.length` BEFORE popping to isolate the current layer.

Gemini Notebook

The Binary Search Tree (BST) Exploit



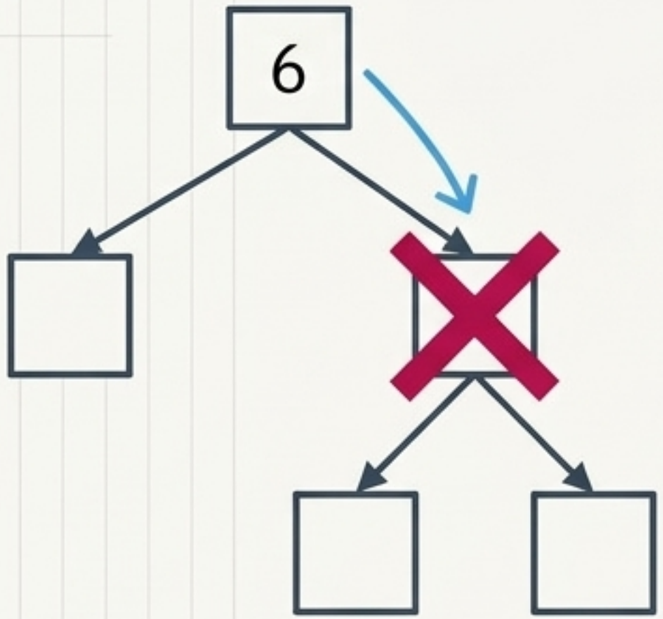
USE CASES: Validating BSTs, K-th Smallest Element, O(log N) searches.

INVARIANT: Left < Node < Right. **In-order** traversal of a valid BST **ALWAYS** yields a **strictly increasing** sequence.

Gemini Notebook

Spotlight: Validating a BST (Medium)

`prev = null`



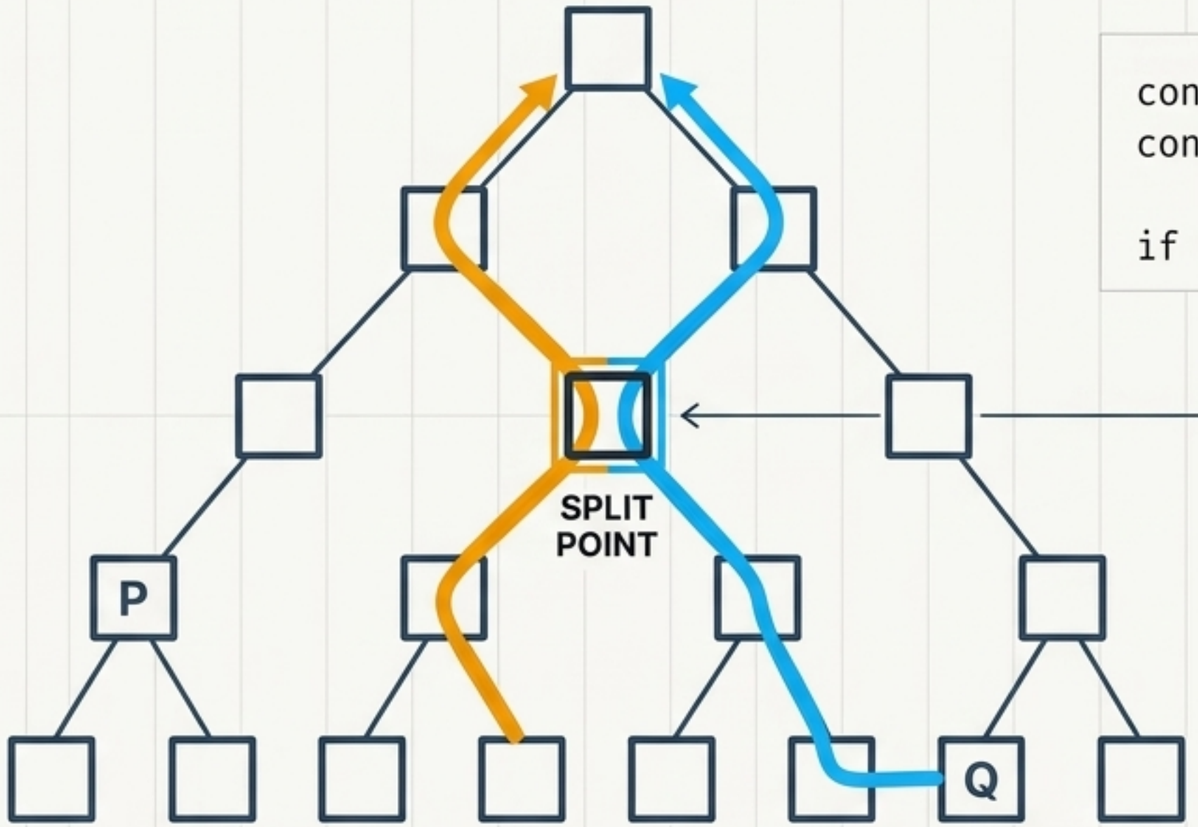

```
if (prev !== null && prev >= node.val)
  return false;
```

Duplicate Trap: Using > allows duplicates. Must be strictly greater (>=).

Integer Overflow Trap: Initializing prev = -Infinity fails if the tree contains the minimum 32-bit integer. Use null.

Gemini Notebook

Lowest Common Ancestor (The Protocol)




```
const left = dfs(node.left, p, q);
const right = dfs(node.right, p, q);

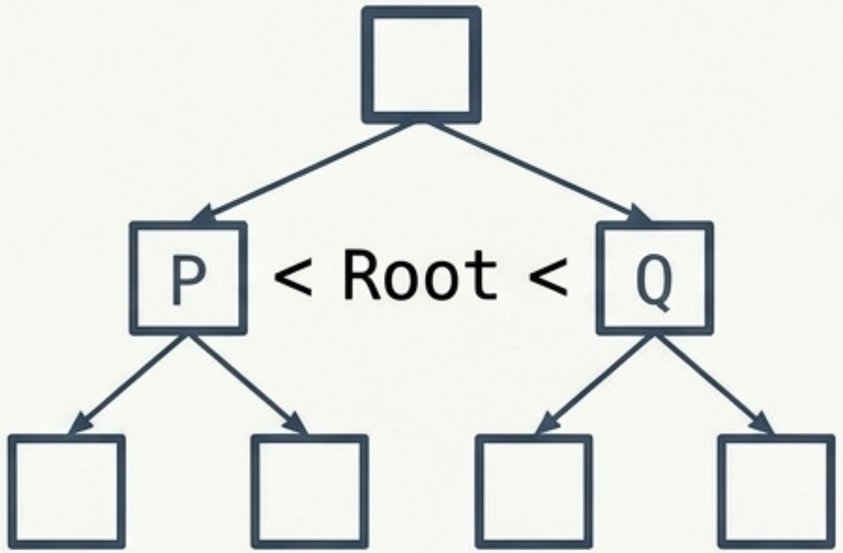
if (left && right) return node; // The Split Point
```

INVARIANT: If a node receives non-null signals from BOTH **left** and **right**, it is the lowest common ancestor.

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Spotlight: LCA in a BST vs. Distance Formula

BST Optimization



Drops to O(log N) by eliminating branches.

Distance Math

```
Dist(U, V) =
    Dist(Root, U) +
    Dist(Root, V) -
   (2 * Dist(Root, LCA))
```

INVARIANT: Finding the LCA is the **skeleton key** for solving all **tree-distance** and **shortest-path** problems.

Gemini Notebook

L4/L5 Interview Survival: Traps & Pitfalls

Space Complexity Ignorance

Always specify `O(H)` recursion stack space (Balanced `O(log N)` vs Skewed `O(N)`).

JS/TS Queue Atrocities

Using `Array.shift()` makes BFS O(N^2) in JavaScript. Must use a head index pointer (`queue[head++]`) to simulate O(1) shifts.

Pointer Loss

Mutating links (e.g., inverting) without caching temporary references (`const temp = node.left`).

The Null Root

Forgetting `if (!node) return` as the mandatory first line of recursive helpers.

Gemini Notebook

Summary: The 30-Second Mental Checklist

The Prompts	The Instants (Mental Triggers)
☑ Need Max Depth / Balance?	→ **Bottom-Up DFS (Post-Order).** Aggregate.
☑ Need Shortest Path / Levels?	→ **Queue BFS + len(queue)** Snapshot.
☑ Need Path Sum / Targets?	→ **Top-Down DFS (Pre-Order).** Backtrack state.
☑ Need to Validate / Sort?	→ **In-Order Traversal.** Track prev.
☑ Need Common Parent?	→ **Post-Order Bubble.** Look for left && right.

The Algorithmic Blueprint - End of Playbook.

Gemini Notebook